Machine Learning: k-Means and PCA for Regime Detection

Zhidian Jing, Siqi Zhang, Stefaan Fernando, Daniel Greenstein, Oscar Peyron
University College London – COMP0040

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1. Abstract

In this project, we apply k-means clustering and PCA to solve the problem of market regime detection in multi-dimensional financial time-series. We compare these techniques against other regime detection methods: the Hidden Markov Model with Gaussian Mixtures (GMMHMM), sliced-Wasserstein k-means clustering, and a simple peak-to-trough-based regime labelling method. We evaluate the quality of each detection method using a trading strategy that goes long when the label indicates a bullish regime and flat otherwise, comparing cumulative returns across methods and regions (America, Europe, Asia).

2. Data Processing and Feature Engineering

We track 9 major stock indices across 3 regions to develop our analysis.

Table 1: Index universe by region.

Region	Index	Ticker
America	S&P 500 / Dow Jones / Nasdaq	`^GSPC`, `^DJI`, `^IXIC`
Europe	FTSE 100 / CAC 40 / DAX	`^FTSE`, `^FCHI`, `^GDAXI`
Asia	KOSPI / Nikkei 225 / Hang Seng	`^KS11`, `^N225`, `^HSI`

2.1. Raw Data Structure

Each index is downloaded from Yahoo Finance with daily OHLCV columns. After merging, the full DataFrame uses a MultiIndex on columns of the form `(index_name, OHLCV_field)`.

2.2. Feature Engineering

From the closing prices we compute the following features for each index:

Log-returns. Over multiple horizons (1, 5, and 21 trading days):

$$r_{t,p} = \ln\left(\frac{P_t}{P_{t-p}}\right). \tag{1}$$

Rolling annualised volatility. With windows of 21 and 63 days:

$$\sigma_{t,w} = \mathrm{std}(r_{t-w:t}) \times \sqrt{252}\,. \tag{2}$$

Rolling correlation. For each pair of indices X and Y at time t with window w:

$$\rho_{X,Y,t} = \frac{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)(r_{Y,t-i} - \bar{r}_Y)}{\sqrt{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)^2 \ \sum_{i=0}^{w-1}(r_{Y,t-i} - \bar{r}_Y)^2}} \tag{3}$$

Momentum Z-score. Measures mean-reversion by normalising distance to the moving average:

$$Z_{t,w} = \frac{P_t - \mathrm{MA}_{t,w}}{\sigma_{t,w}} \tag{4}$$

RSI. The Relative Strength Index captures overbought and oversold conditions:

$$\mathrm{RSI}_t = 100 - \frac{100}{1 + \frac{\text{Average Gain}_t}{\text{Average Loss}_t}} \tag{5}$$

where Average Gain_t and Average Loss_t are computed over 14 days.

Building the feature set. We merge all features (log-returns, rolling volatility, correlation, z-score, RSI) into a single DataFrame. Per index: 3 log-returns + 2 volatilities + 1 z-score + 1 RSI = 7 features; across 3 indices this gives 21 features, plus 3 pairwise correlations, yielding a raw feature matrix of 24 columns. The number of rows equals the trading days minus a 63-day warm-up period. We standardise features and apply PCA retaining 95% of variance.

Table 2: Feature set dimensions after PCA (95% variance retained).

Region	Raw dimensions	After PCA
America	$(3460, 24)$	$(3460, 7)$
Europe	$(3523, 24)$	$(3523, 10)$
Asia	$(3561, 24)$	$(3561, 14)$

2.3. Pipeline Summary

Yahoo Finance API → Raw OHLCV → Merged DataFrame → Close prices

↓

Feature Engineering (returns, volatility, correlation, z-score, RSI)

↓

Standardisation → **PCA** → **K-Means Clustering**

Figure 1: Data pipeline from raw download to regime labels.

3. Mathematical Foundations

3.1. K-Means Clustering

K-means partitions data into K clusters by minimising the within-cluster sum of squares (WCSS):

$$\mathrm{WCSS} = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 \tag{6}$$

where C_k is the set of points in cluster k and μ_k is its centroid. In high-dimensional settings, the curse of dimensionality can limit performance; we therefore reduce dimensionality via PCA before clustering.

3.2. Principal Component Analysis (PCA)

PCA finds directions of maximum variance via the singular value decomposition of the centred data matrix:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \tag{7}$$

where $\mathbf{U}$ contains left singular vectors, $\mathbf{\Sigma}$ is a diagonal matrix of singular values, and $\mathbf{V}$ contains the principal components. We retain the top d^* components capturing $\geq 95\%$ of variance.

3.3. Alternative Regime Detection Methods

To provide a broader comparison, we also evaluate several benchmark regime-detection methods with different modelling assumptions.

3.3.1 Gaussian-Mixture Hidden Markov Model (GMM-HMM)

The Hidden Markov Model assumes that the market evolves through a sequence of unobserved states, with state transitions governed by a first-order Markov chain:

$$P(z_t = j \mid z_{t-1} = i) = A_{ij} \tag{8}$$

where A_{ij} denotes the probability of moving from hidden state i to hidden state j. Conditional on the hidden state, the observed feature vector x_t is modelled as a Gaussian mixture:

$$p(x_t \mid z_t = k) = \sum_{m=1}^{M} \pi_{k,m} \mathcal{N}(x_t \mid \mu_{k,m}, \Sigma_{k,m}) \tag{9}$$

where $\pi_{k,m}$ are mixture weights, and $\mu_{k,m}, \Sigma_{k,m}$ are the corresponding mean vectors and covariance matrices. This allows the model to capture both temporal persistence in market regimes and more flexible within-state feature distributions.

3.3.2 Sliced-Wasserstein K-Means

Unlike standard K-means, sliced-Wasserstein K-means compares rolling return windows through their empirical distributions rather than Euclidean distances between daily feature vectors. Given two distributions μ and ν, the sliced-Wasserstein distance is defined as

$$SW_p(\mu, \nu) = \left(\int_{\mathbb{S}^{d-1}} W_p^p(P_\theta \# \mu,\ P_\theta \# \nu)\, d\sigma(\theta) \right)^{1/p} \tag{10}$$

where P_θ denotes projection onto direction θ, $P_\theta \# \mu$ is the projected distribution, and W_p is the one-dimensional Wasserstein distance. In practice, this is approximated using a finite number of random projections. This method is therefore more sensitive to changes in distributional shape and tail-risk structure across rolling windows.

3.3.3 Peak-to-Trough and Long-Only Benchmarks

As a simple rule-based benchmark, peak-to-trough labelling identifies crisis periods using drawdowns from the running peak of the cumulative return series:

$$DD_t = \frac{C_t - \max_{s \le t} C_s}{\max_{s \le t} C_s} \tag{11}$$

where C_t denotes the cumulative portfolio value at time t. A period is labelled as a crisis regime whenever DD_t falls below a chosen threshold. Finally, a long-only strategy is included as a passive benchmark that remains fully invested throughout the sample period.

4. Proposed Algorithm

Algorithm 1 K-Means Regime Detection Pipeline

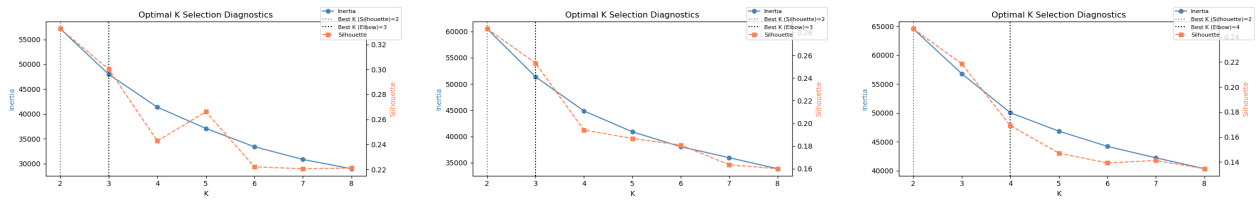
Require: Closing prices $\mathbf{P} \in \mathbb{R}^{T \times N}$, variance threshold α, range $[K_{\min}, K_{\max}]$
Ensure: Regime labels $\ell_t \in \{0, \ldots, K^* - 1\}$, ordered by mean return

— **Stage 1: Feature Engineering** —
for each index $n = 1, \ldots, N$ **do**
  Compute log-returns $r_{t,p}^{(n)}$ for $p \in \{1, 5, 21\}$
  Compute rolling volatility $\sigma_{t,w}^{(n)}$ for $w \in \{21, 63\}$
  Compute momentum z-score $Z_t^{(n)}$ and $\mathrm{RSI}_t^{(n)}$
end for
for each pair (i, j) with $i < j$ **do**
  Compute $\rho_t^{(i,j)} \leftarrow \mathrm{RollingCorr}_{63}(r^{(i)}, r^{(j)})$
end for
Concatenate into $\mathbf{F} \in \mathbb{R}^{T' \times D}$; drop warm-up rows
— **Stage 2: Standardisation & PCA** —
Z-score normalise each feature; compute SVD of $\tilde{\mathbf{F}}$
Select $d^* = \min\{d : \sum_{i=1}^{d} \sigma_i^2 / \sum_{i=1}^{D} \sigma_i^2 \geq \alpha\}$; project $\mathbf{X} \leftarrow \tilde{\mathbf{F}}\, \mathbf{V}_{:,1:d^*}$
— **Stage 3: Optimal K & Clustering** —
for $K = K_{\min}, \ldots, K_{\max}$ **do**
  Run K-Means; record inertia $\mathcal{I}(K)$ and silhouette $\mathcal{S}(K)$
end for
$K^* \leftarrow \arg\max_K \mathcal{S}(K)$; run final K-Means with K^*
— **Stage 4: Regime Ordering** —
Compute mean basket return $\bar{r}_k$ per cluster; relabel 0 = bearish, $K^* - 1$ = bullish
return $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_{T'})$

5. Results

5.1. Optimal K Selection

We applied k-means for $K \in \{2, \ldots, 10\}$ on the PCA-reduced feature space and computed inertia and silhouette scores. As shown in Figure 2, the silhouette score is maximised at $K = 2$ across all regions, suggesting two dominant regimes over 2006–2020.This result suggests that, given the sample period and feature design adopted in this study, market states are primarily characterised by two dominant structures: a relatively weak bearish regime and a relatively strong bullish regime. Relative to larger values of K, the two-regime specification achieves better cluster separation while avoiding the introduction of excessive noise and potential overfitting. It also preserves the stability and interpretability of the detected regimes.

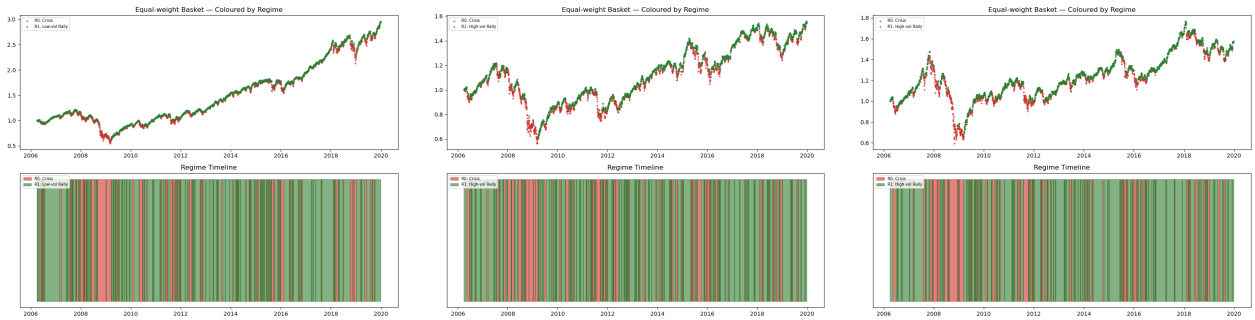


(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 2: Inertia and silhouette scores for different K values across regional indices.

5.2. Regime Detection Results

Figure 3 shows the detected regimes overlaid on an equally weighted basket of indices for each region. In the figure, the red segments represent the crisis regime identified by the model, while the green segments indicate more favourable regimes. These periods broadly align with major historical episodes of market stress, including the 2008 global financial crisis, the 2011 European debt crisis, and the 2015–2016 period of heightened market volatility.

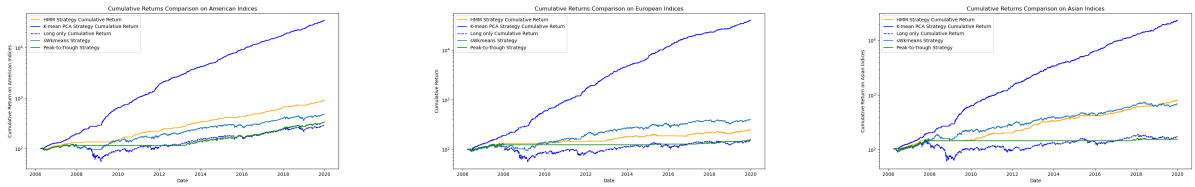


(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 3: Regime detection on equally weighted baskets for each region.

5.3. Trading Strategy Comparison

To evaluate regime quality, we implement a long/flat strategy: go long in bullish regimes, hold cash otherwise. We benchmark PCA K-Means against GMMHMM, sliced-Wasserstein k-means, peak-to-trough labelling (20% drawdown threshold), and a long-only baseline. All strategies start with $100 over March 2006 – January 2020.



(a) American indices. **(b)** European indices. **(c)** Asian indices.

Figure 4: Cumulative returns of trading strategies based on different regime detection methods.

The PCA K-Means strategy significantly outperforms all alternatives across every region, as summarised in Table 3.

Table 3: Backtest performance summary across regional indices.

Region	Metric	PCA K-Means	HMM	sWK-Means	Peak-to-Trough	Long Only
American	Cumulative Return (%)	34,808.39	915.48	483.31	337.18	293.05
	Annualised Return (%)	53	16	12	9	8
	Annualised Volatility (%)	10	6	15	10	19
	Sharpe Ratio	4.37	2.74	0.85	0.95	0.51
European	Cumulative Return (%)	39,532.73	247.17	401.41	155.08	149.26
	Annualised Return (%)	53	7	10	3	3
	Annualised Volatility (%)	11	7	15	6	20
	Sharpe Ratio	3.91	0.94	0.72	0.57	0.24
Asian	Cumulative Return (%)	23,225.90	810.79	687.00	155.36	173.54
	Annualised Return (%)	47	15	15	3	4
	Annualised Volatility (%)	10	8	14	6	19
	Sharpe Ratio	3.76	1.90	1.04	0.54	0.30

Looking at the performance metrics, the PCA K-Means strategy achieves annualised returns of 47–53% with Sharpe ratios above 3.7 across all regions, while alternatives lag significantly behind. The long-only benchmark delivers much lower returns and Sharpe ratios. We can therefore see that PCA K-means capture market regimes in a highly granuar way, as opposed to other methods which are either too noisy (sW k-means) or too coarse (peak-to-trough). The HMM performs decently but still falls short of the PCA K-Means approach, likely due to its parametric assumptions and sensitivity to regime duration. However, while we present performance metrics for trading strategies, those results have to be taken as a descriptive accuracy rather than a predictive accuracy, as for all regime detection methods, we have high look-ahead bias when labelling regimes. This is because we

are using the entire dataset to detect regimes, and then backtesting a strategy on the same dataset. Therefore, while the PCA K-Means method shows superior performance in this backtest, further work would be needed to evaluate its predictive power in an out-of-sample context.

Method-level differences. From a modelling perspective, these performance differences are consistent with the mechanisms of the methods. PCA K-Means clusters daily observations in a denoised, PCA-reduced feature space built from returns, volatility, correlations, momentum z-scores, and RSI, and the resulting regimes are then ordered by mean return. This makes the method particularly effective at separating market states that are directly relevant for trading. By contrast, GMM-HMM uses the same feature set but imposes a latent-state structure with probabilistic transitions, allowing it to capture regime persistence over time at the cost of stronger modelling assumptions. Sliced-Wasserstein K-Means instead clusters rolling return windows based on distributional shape and tail-risk structure, which makes it more sensitive to local changes but also more prone to noisy regime switching. Peak-to-trough labelling is more interpretable but relies only on drawdown information, while the long-only strategy provides a passive benchmark with no regime adaptation.

Regional differences. The regional results in Table 3 also highlight meaningful differences across markets. In the American indices, the advantage of PCA K-Means is likely supported by the strong co-movement among the S&P 500, Dow Jones, and Nasdaq, which allows PCA to extract common market structure effectively; HMM also performs relatively well here, suggesting clearer regime persistence. In the European indices, PCA K-Means achieves the largest relative advantage over the benchmarks, which is consistent with the more pronounced structural stress episodes in the sample, particularly the 2011 European debt crisis. In the Asian indices, although PCA K-Means remains the strongest method, the gap relative to HMM and sWK-Means is narrower. This is consistent with the higher retained PCA dimensionality reported earlier, suggesting a more heterogeneous market structure across the KOSPI, Nikkei 225, and Hang Seng, where alternative methods can still capture part of the regime variation.

5.4. Hyperparameter tuning

To further validate the PCA K-Means specification, we performed a grid search over several key hyperparameters for each regional dataset. Specifically, we varied the number of regimes, whether PCA was applied, the retained PCA variance, and alternative return and volatility horizons. The hyperparameter grid is shown in Table 4.

Table 4: Hyperparameter grid for PCA K-Means tuning.

Hyperparameter	Values
$n_regimes$	$\{2, 3, 4, 5\}$
`use_pca`	{`True`, `False`}
`pca_variance`	$\{0.90, 0.95\}$
`return_periods`	{[1,5,21], [1,5,21,63]}
`vol_windows`	{[21,63], [10,21,63]}

Each specification was evaluated using both the silhouette score and the Sharpe ratio of the corresponding regime-based trading strategy. The silhouette score measures how clearly separated the detected clusters are, while the Sharpe ratio measures the risk-adjusted return of the resulting strategy. To balance clustering quality and trading usefulness, we defined a combined score as an equally weighted average of the silhouette score and a rescaled Sharpe ratio:

$$\text{Combined Score} = 0.5 \times \text{Silhouette} + 0.5 \times \frac{\text{Sharpe}}{3}. \tag{12}$$

Table 5: Best hyperparameter configuration by region.

Region	K	PCA	PCA Var.	Return Periods	Vol Windows	Silhouette	Sharpe	Combined
America	2	True	0.90	[1,5,21]	[21,63]	0.3227	4.5167	0.9141
Europe	2	True	0.90	[1,5,21]	[21,63]	0.2754	4.1912	0.8362
Asia	2	True	0.90	[1,5,21]	[21,63]	0.2029	4.3604	0.8282

The tuning results are remarkably consistent across regions. The detailed tuning results for each region are reported in the Appendix. As shown in Table 5, the best-performing specification is the same in all three markets: $K = 2$, with PCA enabled, 90% retained variance, return periods of $[1, 5, 21]$, and volatility windows of $[21, 63]$. This strongly supports the main model specification and confirms that a simple two-regime structure is most effective under the current feature design. The consistency of the top-ranked configuration also suggests that PCA provides a useful, though moderate, improvement. In all three markets, the best non-PCA specification ranks just below the top PCA-based models, indicating that the raw feature set is already informative, while PCA mainly helps by removing redundant variation and producing a more compact representation. Similarly, retaining 90% of variance slightly outperforms retaining 95%, which suggests that a more aggressive dimensionality reduction can reduce noise without losing the key structure needed for clustering.

The tuning results also highlight meaningful regional differences. The American market achieves the highest silhouette and combined score, indicating the clearest regime separation and the strongest overall clustering structure. Europe ranks second, while Asia has the lowest silhouette score, although its Sharpe ratio remains strong. This pattern is consistent with the earlier interpretation that the Asian indices are more heterogeneous, making their regimes harder to separate cleanly in feature space, while the stronger co-movement in the American indices leads to more clearly defined market states.

Adding a 63-day return horizon or a 10-day volatility window does not improve the best result. This suggests that the baseline feature set already captures the main regime information, while additional short- or long-horizon features are more likely to introduce complexity than to improve performance.

6. Conclusion

We have presented a PCA-enhanced k-means clustering pipeline for market regime detection across American, European, and Asian equity indices. The method consistently identifies two dominant regimes (bullish and bearish) that align well with known periods of market stress. When used to drive a simple long/flat trading strategy, the PCA K-Means approach delivers substantially higher cumulative returns and Sharpe ratios than alternative methods (GMMHMM, sliced-Wasserstein k-means, peak-to-trough labelling) and the long-only benchmark across all three regions. These results suggest that combining dimensionality reduction with unsupervised clustering provides a robust framework for capturing market dynamics. Additionally ...

Future work could explore out-of-sample regime prediction, rolling re-estimation windows, and the inclusion of additional asset classes to further validate the approach.

7. Team Work and Contributions

7.1. Individual Contribution

- Zhidian Jing:
- Siqi Zhang:
- Stefaan Fernando:
- Daniel Greenstein:

- Oscar Peyron: Implemented the Data Processing and Feature Engineering Pipeline. Implemented the PCA and K-means clustering framework.

7.2. Reflection on Team Work

A. Detailed Hyperparameter Tuning Results

This appendix reports the top 10 hyperparameter configurations for each regional dataset, ranked by combined score.

A.1. American Market

Table 6: Top 10 hyperparameter configurations for the American dataset.

Rank	K	PCA	PCA Var.	Return Periods	Vol Windows	Silhouette	Sharpe	Combined
1	2	True	0.90	[1,5,21]	[21,63]	0.3227	4.5167	0.9141
2	2	True	0.95	[1,5,21]	[21,63]	0.3091	4.5008	0.9047
3	2	False	N/A	[1,5,21]	[21,63]	0.2947	4.5153	0.8999
4	2	True	0.90	[1,5,21]	[10,21,63]	0.3314	4.1693	0.8606
5	2	True	0.95	[1,5,21]	[10,21,63]	0.3148	4.1617	0.8510
6	2	True	0.90	[1,5,21,63]	[21,63]	0.3286	4.0854	0.8452
7	2	False	N/A	[1,5,21]	[10,21,63]	0.3030	4.1617	0.8451
8	2	True	0.95	[1,5,21,63]	[21,63]	0.3155	4.0989	0.8409
9	2	False	N/A	[1,5,21,63]	[21,63]	0.3018	4.0989	0.8341
10	3	True	0.90	[1,5,21]	[21,63]	0.3191	3.6943	0.7753

A.2. European Market

Table 7: Top 10 hyperparameter configurations for the European dataset.

Rank	K	PCA	PCA Var.	Return Periods	Vol Windows	Silhouette	Sharpe	Combined
1	2	True	0.90	[1,5,21]	[21,63]	0.2754	4.1912	0.8362
2	2	True	0.95	[1,5,21]	[21,63]	0.2618	4.1837	0.8282
3	2	False	N/A	[1,5,21]	[21,63]	0.2508	4.1650	0.8196
4	3	True	0.90	[1,5,21,63]	[10,21,63]	0.2498	3.8258	0.7625
5	3	True	0.95	[1,5,21,63]	[10,21,63]	0.2319	3.7807	0.7461
6	2	True	0.90	[1,5,21,63]	[21,63]	0.2818	3.6083	0.7423
7	3	False	N/A	[1,5,21,63]	[10,21,63]	0.2179	3.7891	0.7405
8	2	True	0.95	[1,5,21,63]	[21,63]	0.2691	3.6115	0.7365
9	2	False	N/A	[1,5,21,63]	[21,63]	0.2562	3.6381	0.7344
10	4	True	0.90	[1,5,21]	[10,21,63]	0.2040	3.7728	0.7308

A.3. Asian Market

Table 8: Top 10 hyperparameter configurations for the Asian dataset.

Rank	K	PCA	PCA Var.	Return Periods	Vol Windows	Silhouette	Sharpe	Combined
1	2	True	0.90	[1,5,21]	[21,63]	0.2029	4.3604	0.8282
2	2	True	0.95	[1,5,21]	[21,63]	0.1922	4.3670	0.8240
3	2	False	N/A	[1,5,21]	[21,63]	0.1845	4.3743	0.8213
4	2	True	0.90	[1,5,21,63]	[21,63]	0.2089	3.7526	0.7299
5	2	True	0.95	[1,5,21,63]	[21,63]	0.1997	3.7529	0.7254
6	2	False	N/A	[1,5,21,63]	[21,63]	0.1911	3.7575	0.7218
7	3	True	0.90	[1,5,21]	[21,63]	0.1689	3.7204	0.7045
8	3	True	0.95	[1,5,21]	[21,63]	0.1582	3.7316	0.7011
9	2	True	0.90	[1,5,21]	[10,21,63]	0.2173	3.5439	0.6993
10	2	True	0.95	[1,5,21]	[10,21,63]	0.2059	3.5613	0.6965